

-
-
-
-
-

INFRASTRUCTURE MODULE OVERVIEWS

Version 3.5.2

Product Guide



FRONTISPIECE

Copyright © 1998, JBA Software Products Limited, Product Services Division,
Eagle House, The Slough, Studley, Warwickshire, B80 7EN England.

All rights reserved.

No part of the material protected by this copyright may be reproduced or utilised
in any form or by any means, stored in a retrieval system, or transmitted, in any
form, or by any means, electronic, mechanical, photocopying, or otherwise,
without the express prior written consent of JBA Software Products Ltd.

Although every care has been taken in the production of this product information
it cannot be guaranteed that the information contained herein is without error or
omission.

This product information relates to:

Infrastructure Modules Version 3.5.2

Contents

CHAPTER 1 APPLICATION MANAGER OVERVIEW	1-1
PREREQUISITES	1-2
SYSTEM CONVENTIONS	1-3
SYSTEM MANAGER	1-5
ADMINISTRATION FUNCTIONS	1-6
APPLICATION MANAGER	1-8
<i>Menus</i>	1-9
<i>Access Control</i>	1-10
<i>Tasks</i>	1-11
<i>User Profile</i>	1-12
<i>Library Lists</i>	1-13
<i>Application Profile</i>	1-14
<i>Management & Control</i>	1-15
<i>Advanced Menu Functions</i>	1-16
CHAPTER 2 HOUSEKEEPING OVERVIEW	2-1
INTRODUCTION	2-2
CHAPTER 3 MACHINE MANAGER OVERVIEW	3-1
INTRODUCTION	3-2
CHAPTER 4 NETWORK MANAGER OVERVIEW	4-1
INTRODUCTION	4-2
CHAPTER 5 EDI APPLICATION INTERFACE OVERVIEW	5-1
INTRODUCTION	5-2
OVERVIEW	5-3
PREMENOS EDI	5-4
RELATIONSHIP TO OTHER APPLICATIONS	5-6
APPLICATION CONFIGURATION	5-7
PROCESSING CYCLE	5-8
GLOSSARY	I
GLOSSARY	II
INDEX	A
INDEX	B

-
-
-
-
-

Glossary



Glossary

-
-
-
-
-

Index



Index

Error! No index entries found.

-
-
-
-
-

Chapter 1

Application Manager Overview



Prerequisites

You should be familiar with your display station and its controls. There are also some elements of its display and certain keys on the keyboard which are standard, regardless of the software which is currently running at the display station, or which hardware system the display is attached to. Some of these keys are:

- Cursor movement keys
- Function keys
- Field Exit keys
- Insert and Delete keys
- The Error Reset key

You should know how to operate you display station when it is attached to the server and running server software. This requires a knowledge of OS/400 and the Control Language (CL) to do such things as:

- Sign on and sign off the display station
- Interact with displays
- Use Help
- Enter control commands and procedure commands
- Call utilities
- Respond to messages
- The above information is contained in the IBM manual: System Operation: Display Station User's Guide SC21-9744. If you require further information, the following IBM manuals may be consulted:

Information Directory, GC21-9678

Licensed Program's Installation Guide, SC12-9765

System Operations: Operator's Guide, SC21-8082

System Conventions

To avoid the repetition of standard facilities and procedures, certain common routines and functions are referred to by the following conventions:

Signing Off

By entering 90 on the command line of a menu you will sign off the system and return to the initial sign on menu provided by the server's operating software.

Screen Rolling

When the number of lines to be displayed exceeds the capacity of the screen, the rolling facility is available. To execute the roll function press the Shift key and either the Roll Up or Roll Down key, or if the work station does not have Roll keys, use the Page Down or Page Up keys.

Enter Key

In conformance with IBM's SAA standards the **Enter** key is used to update the current record or move to the next screen, for example, in the case of selecting an option.



Note: In some instances where updating the record may give rise to complex consequences you may be asked to confirm the request.

Environment Code

Each software module or application belongs to a specified environment.

For example, you may wish to have a live environment and a test environment for your Financial Systems. The environment code is used to differentiate between these environments.

An application is identified using the keys:

Application ID	e.g. GL General Ledger
Environment	e.g. TST Test Environment
Release	e.g. 03 Version 3



Note: Modules are shipped with a blank environment code.

Selecting Menu Options

All menus and sub-menus used by System Manager and Developer modules enable you to:

- Enter option numbers and then press the **Enter** key
- Move the 'action bar' to the required option and press the **Enter** key. The 'action bar' can be moved by using the Roll keys
- You can go directly to menu options by entering /n. Where n is the option number. If there are sub-menus then you may enter /n/n
- The menus of other System Manager and developer modules may be accessed directly by entering /a. Where a is a single character module ID

- /a Application Manager
- /c Administration Functions
- /f File Migration Utility
- /h Housekeeping
- /i IPG/400
- /m Machine Manager
- /n Network Manager
- /r IRG/400 (Report Generator)
- /v VIEWfinder

Additional Function Key

An additional function key is available from the Application Manager menu.

F22=Select Application

Selection of this function key displays the following screen:

The screenshot displays the 'Application Manager' interface. At the top, it shows 'XA200' on the left, 'Application Manager' in the center, and 'System: STUDB30A' on the right. The 'Main Menu' is a list of options: 1. Maintain Applications, 2. Maintain Library Lists, 3. Maintain Tasks, 4. Maintain Menus, 5. Operational Enquiries, 6. Documentation, 7. Utilities, 8. Execute Application Man, and 9. Exit. Below the menu is a 'Selection or command' field with a cursor. At the bottom, function key shortcuts are listed: F3=Exit, F4=Prom, F10=Command Entry, and F12=Pre. The 'Select Application' screen is overlaid on the right, showing a 'Position List To: _ _ _' field, a prompt 'Type option, press Enter.', and a table of applications. The table has columns 'Op ID Env RI Application Text' and lists 'IN 03 Inventory Management.' and 'GL 03 General Ledger.'. At the bottom of the screen, there are more function key shortcuts: F8=Create Application and F12=Cancel, and the word 'Bottom' in the bottom right corner.

```
XA200                      Application Manager                      System: STUDB30A

Main Menu
1. Maintain Applications
2. Maintain Library Lists
3. Maintain Tasks
4. Maintain Menus
5. Operational Enquiries
6. Documentation
7. Utilities
8. Execute Application Man
9. Exit

Selection or command
==>

F3=Exit      F4=Prom
F10=Command Entry  F12=Pre

Select Application
Position List To: _ _ _
Type option, press Enter.
1=Select Application
Op ID Env RI Application Text
- IN      03 Inventory Management.
- GL      03 General Ledger.

F8=Create Application  F12=Cancel

Bottom
```

This enables you to select a particular application to process.

F8=Create Application

Select this function key to create a new application profile.

System Manager

System Manager is the generic name for the following modules:

- Application Manager
- Machine Manager
- Housekeeping
- Networking

To access System Manager enter the command:



IPGCFP4/STRM400



Note: You may wish to access System Manager when you sign on to the Server. To achieve this change your user profile so that STRM400P is the 'initial program to call'. Enter the command:



CHGUSRPRF USRPRF(uuuuuuuuuuu)INLPGM(IPGCFP4/STRM400P)

Administration Functions

The purpose of the Administration Functions is to maintain those parameters which are common across all System Manager modules.



To access Administration Functions from the Manager 400 main menu either enter 9 in the option field and press **Enter**, or use the Roll keys to move the 'action bar' so that it highlights the Administration Functions module and then press **Enter**.

XA800 Administration Functions System: STUDB30A

Administration Menu

1. Maintain User Profiles
2. Maintain Language Parameters
3. Apply Authorisation Codes
4. Maintain Company Parameters
5. Maintain Soft Options
6. Authorisation Violation Log
7. Exit

Selection or command
==>

F3=ExitF4=PromptF6=MessagesF9=Retrieve

F10=Command EntryF12=PreviousF14=View JobsF24=More Keys

The Administration Functions module of System Manager is a prerequisite for the installation of IPG/400, all other modules within System Manager and the Developer products.



Note: Access to Administration Functions is restricted and is controlled by your System Manager user profile. If you are unable to access Administration Functions contact your server security officer. When shipped, System Manager is set up to allow access to the system security officer (QSECOFR).

The following system-wide parameters can be set up and maintained within Administration Functions.

- Language Parameters - In System Manager, users who speak different languages can each use the same server with display screen constants and messages shown in their native language. This option maintains language descriptions for all the language options which exist in the system.
- Company Parameters - When running application software or your own software you may wish to use a multi-company environment. The Company Parameters option is where company details are set up and maintained.

- Soft Options - Many users have a set of jobs or activities which they want to carry out routinely on the server. Soft options enable you to set up those jobs or activities and run them from anywhere within the System Manager environment.

Application Manager

The Application Manager module provides advanced system integration and easy creation and maintenance of security-conscious menu systems.

For users of standard applications, the product is provided in a ready-to-use state. When the supplied version is loaded it will already be aware of the applications which it has been designed to support.

Application Manager enables you to:

- Interchange between application areas more easily
- Create free-format user-defined menus
- Provide password protection at menu option level
- Provide password protection at function level
- Display current system processing, i.e. which terminal is processing which function
- Add tailor-made functions to user-defined menus
- Work in 'expert mode' of menu operation
- Control job queues and print queues for submitted jobs
- Provide 'exclusive-use-needed' control over applications
- Provide control of the multi-language environment for all standard applications

Application Manager makes it easy to move from one module to another; not only can it provide, on a single menu, a series of options regardless of the module to which they belong, but it can also be used to restrict access at company (for multi-company products), module, menu and menu option level. Application Manager achieves this by providing a simple method of tailor-made menu generation.



Note: Since details from the user profile are used within other System Manager modules, maintenance of user profiles is carried out from the Administration Functions module. Reference to the user profile is made where it is relevant to the Application Manager module.

Menus

Application integration is achieved via the generation of tailor-made menus. Application Manager provides total flexibility in its approach to the building of menus, and thus provides a choice as to the accessibility and content of the user menu.

The user has total flexibility in the layout of his/her menu. Most field-level display attributes may be used (for example, highlighting, underlining, and colour), and the menus are in no way pre-structured as, for example, in two columns. Therefore the user has total freedom to lay out the menu display exactly as he/she wishes.

The use of function keys is totally soft-coded, and may be associated with any action which is required.

Access Control

The accessibility feature controls which companies (for multi-company environments), which applications, and which functions within applications to which a user is granted access. Menu content may be restricted to those functions to which the user is authorised. Access to functions for which the user has not been granted rights of usage will be inhibited at execution time.

This is achieved in two stages:

- At user level, by selecting the menus and menu activities to which the user is authorised.
- At application level, by selecting specific tasks from an application to which the user is allowed access, and within which companies. For example, the user's profile may be set up to enable access to all enquiry menus and all enquiry tasks in a selected application.

Tasks

Each activity which can be carried out under the control of Application Manager is defined as being a task. This describes, for example, which program should be called to perform this activity and whether it is a batch (submitted) or an interactive task. An application may be thought of as a set of tasks, logically grouped together - Accounts Receivable, for example, or Sales Order Processing. To use an entire application, you need to be authorised to all the tasks for that application.

A single menu can feature tasks from many different applications and can thus provide a seamless user environment.

User Profile

In addition to the server user profile, each user has a System Manager user profile for Application Manager. This controls:

- The default sign-on menu to be used.
- The default company assigned (where a multi-company environment is active).
- The default job and print queues. These may be overridden at company and task level. For example, you may wish to have invoices printed on a dedicated printer.
- Whether the user is authorised to enter commands.
- Whether an alternative language is to be used.

In addition, user authorisation to other menus can be granted and authority to application-defined tasks and the companies within applications can be given.

Library Lists

Library lists are set up within the system by reference to a set of standard library list files which are supplied with the system. These library lists are set up to cater for all standard usages of the applications, but you can change them to allow for tailor-made modifications to your own applications. Each application has its own set of library lists. Library lists may be set up to cater for test or training environments where impact on the live data must be eliminated.

Application Profile

Each application must have its own application profile, defining which version of the application is in use and the name of that application. The application profile specifies the default library list to be used, and whether or not a specific application day-end program is required.

Management & Control

Application Manager provides automatic error logging. In the event of an error the system displays a user-friendly message, informing you of an abnormal termination. The system also writes an error record to a file you can enquire on to determine the cause of the error. The error file may also be printed and the results sent to the software support organisation if necessary.

Application Manager enables you to allocate any or all applications for exclusive use, for example for copying files to off-line storage. The system will notify you if anyone is using the application, and permit the sending of a user-defined message to any or all users notifying them of the need for exclusive use of the system. Any user attempting to sign on to an allocated application will receive a message to the effect that the application is not available, and will be unable to access any functions within the application.

Advanced Menu Functions

In addition to providing menus from which the user may enter options or select function keys, you are also able to:

- Use menus in expert mode - If an application has a hierarchical menu structure you may simply enter the name of a menu in the menu option field to go directly to that menu rather than by selecting options from a series of menus to reach the desired menu. In addition, if you know the specific option number required from the menu you can enter this together with the menu name to directly access the task required.
- If you do not know how to access a required task by menu navigation then a search facility is available. This enables you to enter a word contained within the description of the task. A list of tasks which contain the specified word (which you are authorised to) is then displayed, and you may select one of the tasks to process.
- In addition to exact word matches when searching for tasks you are able to set up synonyms to reflect your business. For example, you may wish to have the word 'Item' associated with the words 'Part', 'Product', and 'Assembly'. Once you have set up your own synonyms the task searching facility is used as described above and is transparent to the user.

To access Application Manager enter '4' in the option field and press **Enter**, or use the Roll keys to move the 'action bar' so that Application Manager is highlighted and press **Enter**.

The screenshot shows a terminal window titled 'Application Manager'. At the top left is 'XA200' and at the top right is 'System: STUDB30A'. In the center is a box titled 'Main Menu' containing a list of options: 1. Maintain Applications, 2. Maintain Library Lists, 3. Maintain Tasks, 4. Maintain Menus, 5. Operational Enquiries, 6. Documentation, 7. Utilities, 8. Execute Application Manager, and 9. Exit. Below this box is a prompt 'Selection or command' followed by a line with a cursor. At the bottom, there are four columns of function key definitions: F3=Exit, F10=Command Entry, F4=Browse, F12=Previous, F6=Messages, F14=View Jobs, F9=Retrieve, and F24=More Keys.

```
XA200                               Application Manager                               System: STUDB30A

Main Menu
1. Maintain Applications
2. Maintain Library Lists
3. Maintain Tasks
4. Maintain Menus
5. Operational Enquiries
6. Documentation
7. Utilities
8. Execute Application Manager
9. Exit

Selection or command
=>

F3=Exit      F4=Browse    F6=Messages  F9=Retrieve
F10=Command Entry  F12=Previous F14=View Jobs F24=More Keys
```


-
-
-
-
-

Chapter 2

Housekeeping Overview



Introduction

To select a function in Housekeeping either enter the required option number in the command line or use the Roll keys to move the action bar so that the required function is highlighted, and then press **Enter**.

For each AS/400 computer installation there is a need to ensure the timely saving of the system's libraries and folders to off-line storage. The Housekeeping module provides the tools and utilities to perform and manage these activities.

The user specifies, for each library or folder, the desired save frequency (the number of days required between saves). The process of saving to off-line storage then involves two phases:

In the first phase the libraries which need to be saved are identified. This is done by comparing the date the library was last saved with today's date to calculate the number of elapsed days since the last save was made. The value of elapsed days is then compared with the save frequency as specified above. If a library qualifies for saving on the due date basis (as described) and changes have taken place since it was last saved, then it is saved in this processing run.

There is also a flag to force saving on the next run irrespective of due date and changes made. If you set this flag then the library is saved in this run without any eligibility checks being made.

This phase, the identification of libraries to be included in the 'save set', will normally be scheduled as an overnight task since it may typically take between one and three hours to execute depending on the number of objects held on the system.

The second phase - the actual saving to off-line storage - will differ system by system dependent upon system-level decisions having been made for each AS/400 installation.

The first method of saving is to associate a defined tape, cartridge, or diskette (identified by volume ID) for each library. This information is held on the system - for example, library ABC1234 is stored on tape OS2063. (The tape storage location is also optionally stored on the system along with the tape volume ID) The actual saving to off-line storage by tape, cartridge or diskette involves the system presenting to the user a display of libraries which are to be saved. This display is the 'save set' which was developed in the 'identification phase' as described above. The user must then select each library in turn from the displayed list. After selecting a library from the list, the system will instruct him/her to load the appropriate tape (diskette or cartridge). By merely responding positively to the tape load instruction (simply pressing **Enter**) the system will carry out the save and simultaneously update the control information held by the system. The process is then repeated until all libraries due to be saved on this run have been selected.

The second method involves the saving to disk (either a dedicated UASP or 'normal' ASP) of all the libraries in the save set. This process will typically be performed as an automated 'overnight' job and pre-supposes that there exists nn days (nn is provided at run-time by the user).

In addition to the above facilities which provide timely saving to off-line storage, and the identification of potentially obsolete libraries, the Housekeeping module also provides graphic representation (bar charts) of library size information. These bar charts display how the size of a library has changed over a period of six months or so, thus helping to plan future secondary storage (disk) requirements for the AS/400 installation.

Other key features provided by the Housekeeping module are:

- Creation of libraries
- Archiving libraries prior to deletion
- Media control
- Library compression
- File re-organisation

To access Housekeeping enter either 6 in the option field and press **Enter**, or use the Roll keys to move the 'action bar' so that it highlights the Housekeeping module and then press **Enter**.

```
HK000                      Housekeeping                      System: STUDB30A

Main Menu
1. Control Details
2. Definitions
3. Save to Media
4. Operational Enquiries
5. Documentation
6. Execute Day-End Functions
7. Restore from Media
8. Exit

Selection or command
=> _____

F3=Exit      F4=Prompt    F6=Messages  F9=Retrieve
F10=Command Entry  F12=Previous F14=View Jobs F18=View Prints
```

Alternatively you may wish users to directly access Housekeeping when they sign on to the IBM AS/400. To achieve this change your IBM AS/400 user profile so that STRIPGHKP is the 'Initial program to call'. Enter the command:



CHGUSRPRFUSRPRF(uuuuuuuuuu)INLPGM(IPGCFP4/STRIPGHKP)

-
-
-
-
-

Chapter 3

Machine Manager Overview



Introduction

The Machine Manager module performs tasks which fall into two broad categories:

1 Housekeeping tasks

These are normally performed when all or most of the interactive users have finished with the system, and gone home. Tasks which fall into this category might include the re-organisation of database files, the securing of the off-line storage of the system's programs and data, or the production of some lengthy reports which might, for example, be scheduled to run on a weekly basis.

The time at which these overnight activities are initiated is decided by the user. At any time this value may be modified permanently or altered merely for the current night's trigger time.

Jobs scheduled for overnight execution may be set up and 'masked' to run on any night (or each night) of the week, or to run on specific dates. The 'mask' may be temporarily overridden for a particular night. The facility to schedule the date on which a job runs is also available. For example, you may wish to run a task or set of tasks on the 10th of each month.

After the completion of all tasks which have been scheduled for overnight running, the machine if required will power itself off. This process is perfectly normal and is recommended for the IBM AS/400. On successfully re-starting (IPL) the system will run through a list of tasks which may be similarly scheduled to general use. The re-starting of the IBM AS/400 is fully automated and unattended, the machine is therefore ready for the users when they arrive on the following day.

2 Scheduled tasks

These are those jobs you wish to perform during the normal working day and required their initiation to be scheduled for a specified time. Tasks which fall into this category might include sending a message to a user(s) to remind them a meeting is about to start, the production of a report or initiating an update program during an inactive period, maybe the lunch break.

In addition to the control of the initiation of tasks Machine Manager provides a fast and user friendly method of setting up standard tasks. For example, a task from the Inventory Management application, or a task from the Housekeeping module such as compressing libraries, to be run as part of your day-end routines.

To access Machine Manager enter 6 in the option field and press **Enter**, or use the Roll keys to move the 'action bar' so that Machine Manager is highlighted and press **Enter**.

MM000

Machine Manager

System: STUDB30A

Main Menu

1. Auto Day-Start Jobs

2. Auto Day-End Jobs

3. Job Schedules

4. End Day Times

5. Power Down Times

6. Job Schedule Times

7. Group Schedules

8. Protected Queues

9. Operational Enquiries

10. Exit

Selection or command
=> _____

F3=Exit

F4=Prompt

F6=Messages

F9=Retrieve

F10=Command Entry

F12=Previous

F14=View Jobs

F24=More Keys

-
-
-
-
-

Chapter 4

Network Manager Overview



Introduction

As time goes by every new IBM AS/400 installation will find itself more likely to be included as a member of a communications network. More and more computers are being linked together by advanced communications methods in order to provide more timely management information for their users. This is particularly true of large national and multi-national organisations with many sites, in many instances crossing national boundaries.

The advanced communications features of the IBM AS/400 lend the machine to be a sophisticated role player in this environment.

Network Manager offers a user-friendly interface to the communications network. It provides IBM AS/400 users with the facilities to:

- Send printed reports between IBM AS/400's
- Send objects, folders, and libraries between AS/400's
- Send individual source members between AS/400's in a network
- Initiate display station pass-through from an IBM AS/400 to another system in a complex network

Network Manager also offers the network supervisor a user-friendly interface to the systems which he/she may have to control in his/her network. Network Manager provides facilities to:

- Manage the network, configure the Distribution Services (SNADS), maintain distribution queues and routing entries
- Manage display station pass-through within the user network with soft-coded commands and command prompts. This provides an easy means of specifying the routing to be used when entering display station pass-through from system A to system B in a network
- Monitor the transmission of objects via the network log
- Simplify spool file transfers across peer-to-peer networks, enabling automatic transfer from output queue to output queue on source and target systems

To access Network Manager enter 8 in the option field and press **Enter**, or use the Roll keys to move the action bar so that Network Manager is highlighted and press **Enter**.

NW000 **Network Manager** System: STUDB30A

Main Menu

1. Send Objects
2. Networking
3. Spooled File Distribution
4. Object Distribution
5. Exit

Selection or Command
=> _____

F3=Exit	F4=Prompt	F6=Messages	F9=Retrieve
F10=Command Entry	F12=Previous	F14=View Jobs	F18=View Prints

-
-
-
-
-

Chapter 5

EDI Application Interface Overview



Introduction

Electronic Data Interchange is not a new concept. In the early 1960s many large multi-nationals, such as Philips and General Motors, started to transfer internal information and instructions electronically between different application systems, whether by tape or telecommunications.

Since then, technology and business practice have evolved to the point where EDI is virtually a condition of trade in some business sectors.

The application of EDI has accelerated as companies have come to realise that EDI can bring many measurable benefits:

Speed

EDI sends and receives documents quickly. There is none of the delay associated with mailing the documents.

Accuracy

Since EDI transmits documents electronically, there is no need to re-enter the information, thus reducing keying errors.

Cost Savings

The quick turn-around of documents reduces inventory overhead and increases the available cash flow. EDI also reduces mailing costs.

Productivity

Employees are free to perform different tasks. They are no longer needed to enter documents or address, stamp and mail envelopes.

The EDI solution has two closely integrated components. These are:

- The EDI Application Interface module
- Premenos EDI product

Together, these products satisfy the requirements of a successful EDI installation, namely:

- Controlled Application Interfaces
- Standard Message Format Support
- Conversion Facilities
- Communications Management Facilities

EDI will provide support for the most widely-used Message Standards and Value Added Networks. It will also give mailbox access and document conversion facilities. The Application Interface module supplies the facility to import and export documents - in a controlled fashion - to and from the standard Sales Order Processing and Purchase Management modules.

Overview

EDI Application Interface enables the import and export of documents to and from Sales Order Processing and Purchase Management. These documents, traditionally sent and received in a paper form, can now be exchanged as electronic messages. The application will support:

- Inbound sales orders and order amendments
- Outbound acknowledgements
- Outbound advice of despatch
- Outbound sales invoices and credits
- Outbound purchase orders and amendments
- Outbound vendor schedules
- Outbound catalogues
- Inbound acknowledgements
- Inbound advice of despatch
- Inbound supplier invoices (onto the Accounts Payable log)
- Inbound schedule orders
- Inbound advanced shipment notes

The application can be used in two ways.

The application is designed to work very closely with the Premenos EDI product, EDI/400. This will cater for the conversion of messages into and out of standard message formats. The Application Interface module adds new function for users of Sales Order Processing, Vendor Scheduling and Purchase Management in the shape of direct calls to EDI. So, for example, one interface option will allow the preparation of electronic purchase orders, and will also invoke a routine in EDI to convert those orders into standard message formats for onward transmission to suppliers. This close-coupled link to EDI removes the need for repeated manual intervention in the conversion and transmission process, which can be time consuming, and thus costly.

The second potential use for the interface module is as an electronic batch transaction facility. For example, it is common in certain industries for sales orders to be captured by the field sales force on portable microcomputers. Traditionally, these have then been re-entered onto the central Sales Order Processing application. With the arrival of the Application Interface module, this is no longer necessary. Using existing facilities, the orders can be loaded up onto the host system, and taken into the order bank, in a controlled fashion, using the Application Interface module. There are many similar potential uses for the module.

The control aspect must be emphasised. This module applies the same validation logic to inbound and outbound messages as is used in the conventional operation of the system. Full exception and audit reporting is provided.

Premenos EDI

EDI provides the following functions:

- Reformatting of messages between the private format of the Application Interface module and the formats of the major international EDI Standards
- Communication modules to enable direct and straightforward connection to all the principal value added networks
- Support for point-to-point communications where a value added network is not in use
- Full transaction logging, error recovery and security features

Support Of EDI Standards

EDI will support the following national and international EDI Standards:

- ANSI X12
- INCOME
- EDIFACT
- ODETTE
- Tradacoms
- TDB
- US

In addition, EDI contains a standard modification tool, which will allow modification of standards already defined to the product, and definition of new standards.

Value Added Network Support

EDI is supplied with a module enabling connection to the Premenos Information Network. In addition, further communication modules will be available to enable connection to any, or all of the following value added networks:

- INS/TRIDENT
- AT&T Easylink
- GEIS network services
- McDonnell Douglas network services
- Control Data/Redinet
- Sterling/Ordernet Version 3
- Kleinschmidt network services
- TranSettlement/Transend
- General Motors/GMNET
- Ford/SOLMIS

As well as the value added networks (VAN) listed above, EDI will allow direct point-to-point communications between systems, where a VAN is not used.

Relationship To Other Applications

The following applications are pre-requisite to the EDI Application Interface:

- Inventory Management
- Sales Order Processing (for sales documents only)
- Purchase Management (for purchasing documents only)
- Vendor Scheduling (for vendor schedules documents only)
- Accounts Payable (for supplier invoices)
- Advanced Receiving (for advanced shipment note documents only)
- Application Manager

The application is designed to be able to add translated documents into the mailbox send queue in the EDI application automatically. This requires, however, the trading partner being defined to EDI.

Application Configuration

To set up the additional data required for sending and/or receiving documents via the EDI Application Interface. The main elements to be created are:

- Documents
- Trading partners
- Location references
- Translation maps
- Code translation tables

Documents

All document types to be sent via EDI must be defined in the Application Interface, e.g. purchase order, sales invoice. Within this, the document ID must be specified for each EDI message standard used.

These definitions will be shared by all companies defined to the application.

Trading Partners

All trading partners (customers or suppliers) who are to be communicated with via EDI must be defined in the Application Interface.

Within this, the document types to be sent to each partner must be selected from those defined above. At this point it should be specified if the document is to be printed and/or sent via EDI for this trading partner.

Further details must be defined for the trading partner in the EDI application. These are required to support the document translation and communication functions in EDI.

Location References

Where delivery is required to multiple locations, a set of agreed location reference codes must be maintained in order to establish the correct delivery address.

Translation Maps

A translation map is required for each trading partner/document combination, whether they are sent or received. This is used by EDI to map data sent via EDI from the Application Interface database into the appropriate EDI message standard, and vice-versa for data received.

Code Translation Tables

A code translation table is required to convert codes, e.g., units of measure, sales tax, into the equivalent codes recognised by the appropriate message standard when sending documents.

Conversely, they are used to translate EDI standard codes into codes for receiving documents.

For more details on setting up trading partners see Trading Partners Maintenance.

Processing Cycle

The following example is provided to illustrate how the EDI processing cycle is supported by Premenos products. The example given is that of an incoming sales order. The procedure for outgoing messages is similar.

Although each step can be executed separately, the products can be configured to allow steps 1, 2 and 3 to be combined into one option leaving step 4 to be executed separately.

Step 1 Access Value Added Network

Having agreed with the applicable electronic trading partners which VAN is to be used to exchange messages, you can define it to EDI. This definition tells EDI how the connection to the VAN should be established.

EDI supports binary synchronous communications with all the networks for which communications modules are provided. For Premenos' IN, full SNA communications are supported. The selected protocols can be used over leased or dialled lines. In the case of dialled connections, either autodial or manual dial is supported. During set-up, other parameters concerning communications with the network will be recorded. EDI stores these parameters with the telephone numbers for dialled connections.

For examples in this product information it is assumed that EDI has established the connections between the AS/400 and an IN.

Step 2 Transfer Documents

Once the communications session is started, EDI will use the previously defined parameters to control the conversation with the network. EDI will then collect any messages sitting in the network mailbox and transfer them down to its own mailbox on the AS/400. It will also check to see if there is any outbound mail (e.g. acknowledgements) that should be sent via IN, and if there are it will transfer them up to the network mailbox. IN will then distribute these to the appropriate trading partners. EDI maintains an on-line log containing the details of each communications session.

Once the transfer process is complete, EDI will - if appropriate - disconnect from the network.

The received message will now be sitting in the EDI mailbox. It can be displayed, printed or deleted while it is there. It can also be translated and moved into the Application Interface Module.

Step 3 Translate To In house Format And Files

Before EDI can successfully translate an incoming order out of its EDI format and into the in-house format, a number of definitions must have been set up within EDI:

- **Trading Partner Transaction Type**

This defines a unique business relationship regarding the two trading partners concerned in the transaction of a particular document, in this case an order. EDI uses this information when handling received messages. The information is also used when addressing and sending outbound messages.

EDI also uses the transaction type to identify the translation map that should be used to translate the document (whether inbound or outbound). The transaction type records other information regarding the relationship that has been agreed with a trading partner, such as the network over which the document should be exchanged, the trading partner's network ID, and version and release values for the standard used.

- **Application Interface File Definitions**

The Application Interface files that will receive the incoming order must be defined to EDI as 'user files'. During translation, EDI directs the translated data to these files.

- **Translation Map**

The relationship between the incoming Tradacoms order and the Application Interfaces files has to be defined. To do this, a 'map' must be created. This map lies at the heart of the translation process. EDI moves data contained in the Tradacoms standard order into the corresponding Application Interface files. A map is required for all EDI documents that are to be sent or received. This is true of all EDI standards.

- **Code Translation Tables**

EDI standards will often have pre-defined codes for certain data elements. These must be translated so as to be accepted as valid. Similarly, if our trading partner uses different values (e.g. UOM, delivery address or product codes), they must also be translated into values acceptable to the particular application. Hence code translation tables. EDI uses these tables to translate standard and trading partner-specific codes into values that are acceptable.

The translation process takes all these elements into account. When completed, the translated order will be written to the appropriate Application Interface file.

Step 4 Application Interface

Once EDI has successfully translated the incoming order and placed it in the Application Interface file; an interface option can be executed to place it on the Sales Order Processing order bank. This interface program will perform the same validation upon the incoming orders that is applied normally to orders entered to the system via the keyboard. Most exceptions will cause the order to be suspended for further investigation. In the case of major exceptions, the order will be rejected. Full exception and control reports are provided.

The logic described in this example is applied to all the documents (whether inbound or outbound) supported by EDI and the EDI Application Interface. In the case of outbound documents (e.g. purchase orders), most of the validation will already have been carried out by the application generating the document. For outgoing documents, EDI and the Application Interface examine the validity of the document as an EDI message, deciding whether they should be sent via EDI and/or printed.

